DATA ANALYTICS WITH POWER BI

PROJECT REPORT

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Crime Report Dashboard: Analysis of DC Crime Incidents

Submitted by

Jyoti Kushwaha

Registration No:

12325612

Programme and Section:

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Under the Guidance of

Mrs SAVLEEN KAUR (UID : 18306)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Jyoti Kushwaha bearing Registration Number 12325612 has completed INT374 project titled, **"Crime Analytics Dashboard"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 24-12-2025

DECLARATION

I, Jyoti kushwaha, student of Bachelor of Technology under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 24-12-2025 Signature

Registration No. 12325612 Jyoti

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the completion of this project titled **"Crime Analytics Dashboard"** First and foremost, I would like to thank Savleen kaur, my respected project supervisor, for her invaluable guidance, support, and encouragement. Her expert advice, timely feedback, and motivation helped me stay focused and complete this project efficiently.

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A special thanks to my peers, friends, and family for their constant moral support and encouragement throughout the project.

Lastly, I would like to acknowledge the use of the dataset provided by data.gov which served as the foundation for the analysis conducted in this report.

Savleen Kaur

Registration No.: 12325612

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Drive Link:

https://drive.google.com/file/d/1OY_DH2R_Cbljbp3uJs_zcWaFXkvUOLf8/view?usp=drive_link

1.Introduction

In the contemporary landscape of public administration, data-driven decision-making has become paramount. Law enforcement agencies generate vast quantities of data daily, not only related to crime incidents but also detailing **victim and suspect demographics** such as age, gender, and race. However, without structured analysis and visualization, this data remains underutilized and fails to support effective policy formulation. This project, titled **"Crime Analysis Dashboard,"** aims to transform raw demographic crime data into meaningful, actionable insights.

Using **Microsoft Power BI**, this project analyzes reported crime records containing detailed **victim and suspect demographic information** collected by the police department. The resulting interactive dashboard enables users to explore crime patterns across different demographic groups, identify vulnerable populations, and understand behavioral trends among suspects. Such insights are critical for developing targeted crime prevention strategies and promoting equitable law enforcement practices.

Traditional crime reporting methods often rely on static spreadsheets or PDF documents, which are difficult to interpret and lack interactivity. As a result, decision-makers face several challenges:

• **Limited demographic insight:** Static reports fail to clearly highlight how crime impacts different age groups, genders, and racial categories.
• **Inefficient policy planning:** Without visual demographic breakdowns, it becomes difficult to design targeted awareness programs or preventive interventions.
• **Lack of integrated analysis:** Crime data and demographic attributes are often analyzed separately, preventing holistic understanding of crime dynamics.

This project addresses these challenges by consolidating crime records into a unified analytical model and presenting them through interactive visualizations.

The primary goal of this project is to develop a **Business Intelligence (BI) solution** that provides:

• **Temporal Analysis:** Understanding how demographic crime patterns change over time (yearly and monthly).
• **Demographic Analysis:** Examining victim and suspect distribution across age groups, gender, and race.
• **Comparative Analysis:** Identifying relationships and contrasts between victim and suspect demographics.
• **Categorical Analysis:** Analyzing crime types in relation to demographic characteristics to prioritize preventive measures.

To accomplish these objectives, the following tools and technologies were used:

• **Microsoft Power BI Desktop:** The primary tool for data ingestion, modeling, and visualization.
• **Power Query Editor:** Used for ETL (Extract, Transform, Load) operations such as data cleaning, standardization, and transformation.
• **DAX (Data Analysis Expressions):** Used to create calculated measures, age group classifications, and percentage-based insights.
• **Microsoft Excel/CSV:** The source format of the raw demographic crime data.

2. Scope of Analysis

The scope of this project is focused on analyzing crime incident data from a demographic and geographical perspective using data analytics and visualization techniques. The analysis aims to understand patterns in reported crime incidents by examining the distribution of victims and suspects across different demographic attributes such as age group, gender, race, and person type, along with temporal and district-level variations.

This project specifically evaluates how crime involvement varies across different population segments and locations over time. The scope includes both high-level aggregated analysis and detailed category-wise analysis to provide a comprehensive understanding of crime dynamics. By combining summary-level insights with detailed breakdowns, the dashboard enables stakeholders to identify dominant trends as well as underlying patterns that may not be evident in traditional reports.

The analysis covers multiple dimensions of crime data. These dimensions include temporal attributes such as year and quarter, demographic attributes such as age range, gender, and race, and spatial attributes such as police district descriptions. Additionally, the analysis differentiates between person types, namely victims and suspects, allowing for comparative assessment of crime involvement across different roles.

A key component of the scope is the objective-wise analysis of crime trends. The dashboard evaluates long-term crime trends over multiple years to identify periods of increase or decline in reported incidents. It also examines crime involvement by gender and age range to highlight vulnerable and high-risk demographic groups. Race-based analysis is included to understand representation patterns within reported crime data, while district-wise analysis provides insight into geographical concentrations of crime activity.

The scope also includes categorical analysis of crime types across districts. By analyzing crime categories in relation to district descriptions, the project helps identify which types of crimes are more prevalent in specific areas. This information is valuable for law enforcement agencies and policymakers in designing targeted prevention strategies and allocating resources efficiently.

Interactive analysis forms an important part of the project scope. The dashboard incorporates slicers and filters that allow users to dynamically explore data based on selected criteria such as time period, district, demographic group, and crime category. This interactivity enables focused analysis without modifying the underlying dataset and supports exploratory decision-making.

It is important to note that the scope of this project is limited to descriptive and diagnostic analysis. The dashboard is designed to analyze historical crime data and identify patterns, trends, and disparities within the available dataset. Predictive modeling, crime forecasting, and real-time monitoring are outside the scope of this project and are identified as potential enhancements for future work.

The analysis is based exclusively on officially reported crime data published by the Metropolitan Police Department through an open government data platform. As a result, the findings are limited to reported incidents and may not capture unreported or underreported crimes. The project does not attempt to infer causation but focuses on presenting factual, data-driven insights through visualization.

Overall, the scope of analysis is to provide a structured, interactive, and visually intuitive approach to understanding crime patterns using demographic and district-level data. The project aims to support data-driven decision-making, improve transparency in crime reporting, and demonstrate the practical application of Business Intelligence tools in the public safety domain.

3. Source of Dataset

The data for this project was obtained from Open Data DC, the official data clearinghouse for the District of Columbia Government. The credibility of the data is high as it is directly published by the Metropolitan Police Department (MPD).

To ensure a comprehensive historical analysis, data was fetched from five distinct endpoints, covering the years 2021 through 2025.

Source URLs: https://catalog.data.gov/dataset/police-department-reported-victim-and-suspect-demographic-data

Attribute Name	Description	Data Type
QUARTER_YEAR	The date on which the crime incident was officially reported to the police department.	Date
PERSON_CRIME_CATEGORY	The classification or category of the crime (e.g., Theft, Assault, Robbery, Burglary).Helps in understanding the nature and severity of offenses.	Text
AGE_RANGE	The age of the victim involved in the reported crime.	Text
DISTRICT	A numeric or coded representation of the police district where the crime was reported. Supports district-level comparison and administrative analysis.	Text
DISTRICT_DESCRIPTION	The descriptive name of the police district or jurisdiction.	Text
PERSON_RACE_DESCRIPTION	Indicates the racial or ethnic classification of the person involved in the crime as recorded by law enforcement level).	Text
PERSON_SEX_DESCRIPTION	Specifies the gender of the individual involved in the crime (e.g., Male, Female,	Text

	Unknown)	
PERSON_TYPE	Identifies whether the individual is a **Victim** or a **Suspect**.	Text
PERSON_COUNT	Represents the total number of persons falling under a specific combination of attributes (age range, gender, race, district, and crime category).	Whole Number
DATA_AS_OF	Indicates the date up to which the data is considered accurate and complete	Date
DATA_LOADED_AT	Represents the timestamp when the dataset was loaded or updated in the system.	Date/Time

quarter_year	person_crime_category	age_range	district	district_description	person_race_description	person_sex_description	person_type	person_count	data_as_of	data_loaded_at
10/1/2021 0:00	VANDALISM	18-29	D	MISSION	White	Male	VICTIM	9	10/23/2025 14:00	10/23/2025 14:07
10/1/2024 0:00	VANDALISM	30-39	X	OUT OF SF	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
4/1/2023 0:00	VANDALISM	Unknown	X	OUT OF SF	Unknown	Unknown	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
1/1/2015 0:00	VANDALISM	40-49	A	CENTRAL	American Indian or Alaskan Nat	Male	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
7/1/2019 0:00	BATTERY/OTHER ASSAULT	18-29	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
4/1/2025 0:00	BATTERY/OTHER ASSAULT	18-29	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
1/1/2019 0:00	BATTERY/OTHER ASSAULT	30-39	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
4/1/2022 0:00	VANDALISM	30-39	D	MISSION	White	Male	VICTIM	12	10/23/2025 14:00	10/23/2025 14:07
1/1/2023 0:00	VANDALISM	40-49	D	MISSION	White	Male	VICTIM	6	10/23/2025 14:00	10/23/2025 14:07
7/1/2025 0:00	VANDALISM	50-59	D	MISSION	White	Male	VICTIM	5	10/23/2025 14:00	10/23/2025 14:07
4/1/2018 0:00	VANDALISM	18-29	A	CENTRAL	Asian or Pacific Islander	Female	VICTIM	4	10/23/2025 14:00	10/23/2025 14:07
4/1/2021 0:00	VANDALISM	30-39	A	CENTRAL	Asian or Pacific Islander	Female	VICTIM	5	10/23/2025 14:00	10/23/2025 14:07
7/1/2024 0:00	VANDALISM	40-49	A	CENTRAL	Asian or Pacific Islander	Female	VICTIM	3	10/23/2025 14:00	10/23/2025 14:07
4/1/2023 0:00	BATTERY/OTHER ASSAULT	30-39	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
1/1/2018 0:00	BATTERY/OTHER ASSAULT	40-49	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
4/1/2024 0:00	BATTERY/OTHER ASSAULT	40-49	D	MISSION	Asian or Pacific Islander	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
10/1/2016 0:00	VANDALISM	Unknown	D	MISSION	White	Male	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
1/1/2022 0:00	VANDALISM	18-29	E	NORTHERN	American Indian or Alaskan Nat	Male	VICTIM	3	10/23/2025 14:00	10/23/2025 14:07
7/1/2017 0:00	VANDALISM	18-29	E	NORTHERN	Asian or Pacific Islander	Female	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
10/1/2017 0:00	DOMESTIC VIOLENCE	60+	J	TENDERLOIN	White	Male	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
10/1/2020 0:00	BURGLARY	18-29	E	NORTHERN	White	Female	VICTIM	15	10/23/2025 14:00	10/23/2025 14:07
1/1/2021 0:00	BURGLARY	30-39	E	NORTHERN	White	Female	VICTIM	22	10/23/2025 14:00	10/23/2025 14:07
4/1/2022 0:00	BURGLARY	40-49	E	NORTHERN	White	Female	VICTIM	6	10/23/2025 14:00	10/23/2025 14:07
1/1/2021 0:00	DOMESTIC VIOLENCE	40-49	C	BAYVIEW	Black	Male	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
10/1/2023 0:00	DOMESTIC VIOLENCE	50-59	C	BAYVIEW	Black	Male	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
7/1/2016 0:00	DOMESTIC VIOLENCE	40-49	UNK	UNKNOWN	Hispanic or Latino	Female	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
7/1/2015 0:00	DOMESTIC VIOLENCE	30-39	X	OUT OF SF	Asian or Pacific Islander	Female	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
4/1/2023 0:00	DOMESTIC VIOLENCE	0-17	X	OUT OF SF	Black	Female	VICTIM	1	10/23/2025 14:00	10/23/2025 14:07
1/1/2021 0:00	AGGRAVATED ASSAULT	18-29	X	OUT OF SF	Unknown	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07

Police_Department_Reported_Vict

Figure 1

	A	B	C	D	E	F	G	H	I	J	K
249425	4/1/2025 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Female	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249426	7/1/2015 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249427	1/1/2020 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Black	Male	SUSPECT	2	10/23/2025 14:00	10/23/2025 14:07
249428	7/1/2023 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249429	4/1/2016 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249430	10/1/2019 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249431	7/1/2025 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249432	7/1/2017 0:00	AGGRAVATED ASSAULT	40-49	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249433	10/1/2019 0:00	AGGRAVATED ASSAULT	50-59	A	CENTRAL	Black	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249434	4/1/2021 0:00	AGGRAVATED ASSAULT	60+	A	CENTRAL	Black	Male	SUSPECT	2	10/23/2025 14:00	10/23/2025 14:07
249435	1/1/2014 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	6	10/23/2025 14:00	10/23/2025 14:07
249436	1/1/2016 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	32	10/23/2025 14:00	10/23/2025 14:07
249437	1/1/2018 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	8	10/23/2025 14:00	10/23/2025 14:07
249438	1/1/2020 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	8	10/23/2025 14:00	10/23/2025 14:07
249439	1/1/2022 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	22	10/23/2025 14:00	10/23/2025 14:07
249440	1/1/2024 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Black	Male	SUSPECT	9	10/23/2025 14:00	10/23/2025 14:07
249441	1/1/2025 0:00	AGGRAVATED ASSAULT	60+	A	CENTRAL	Black	OTHERS	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249442	7/1/2016 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	2	10/23/2025 14:00	10/23/2025 14:07
249443	4/1/2021 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249444	1/1/2014 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	2	10/23/2025 14:00	10/23/2025 14:07
249445	1/1/2016 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	5	10/23/2025 14:00	10/23/2025 14:07
249446	7/1/2018 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249447	10/1/2021 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249448	7/1/2024 0:00	AGGRAVATED ASSAULT	Unknown	A	CENTRAL	Hispanic or Latino	Female	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249449	1/1/2013 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	2	10/23/2025 14:00	10/23/2025 14:07
249450	10/1/2017 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249451	7/1/2023 0:00	AGGRAVATED ASSAULT	18-29	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249452	1/1/2019 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249453	1/1/2023 0:00	AGGRAVATED ASSAULT	30-39	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07
249454	10/1/2022 0:00	AGGRAVATED ASSAULT	40-49	A	CENTRAL	Hispanic or Latino	Male	SUSPECT	1	10/23/2025 14:00	10/23/2025 14:07

Police_Department_Reported_Vict

Figure 2

1 Dataset Preprocessing

After importing the data, several cleaning and standardization operations were performed to ensure consistency and analytical accuracy:

1. **Type Casting:**
   - The **quarter_year** column was maintained as a categorical time field.
   - The **person_count** column was converted to a numeric data type to enable aggregation and KPI calculations.
   - The **data_as_of** and data_loaded_at columns were converted to Date and Date/Time formats respectively.
2. **Null and Missing Value Handling:**
   - Records with missing values in critical fields such as `district`, **person_type**, or **person_crime_category** were filtered out.
   - Non-informative or "Unknown" values in demographic fields were retained only where analytically relevant.
3. **Standardization:**
   - Gender values in `person_sex_description` were standardized (e.g., "M", "Male" → "Male").
   - Consistency was ensured across race descriptions in `person_race_description`.
   - Uniform naming conventions were applied to crime categories and district descriptions.
4. **Data Validation:**
   - Cross-verification was performed to ensure that `person_count` values aligned correctly with the corresponding demographic and district attributes.

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2 Analysis on Dataset (Objective-wise)

4.1 General Description

The final dashboard presents a **high-level executive overview of victim and suspect demographic crime patterns**. A clean and dark-themed layout was used to enhance visual clarity and highlight key trends across different demographic dimensions.

The dashboard is fully interactive—users can filter data by **quarter, district, person type (Victim or Suspect), age range, gender, race, and crime category**. Selecting a specific time period or district dynamically updates all visual components, enabling focused and comparative analysis.

This interactive design allows stakeholders to:

- Identify demographic groups most affected by crime,
- Compare victim and suspect profiles,
- Analyze district-wise crime distribution,
- Support data-driven policy and preventive decision-making.

Figure 3

4.2 Specific Requirements

The analysis was driven by seven specific objectives aimed at uncovering patterns in the data. These objectives cover temporal, spatial, and categorical dimensions of crime.

4.3 Objective 1: Crime Trends over the Years

Objective: To determine the long-term trend of criminal incidents..

Analysis: We aggregated the total count of CCN (Crime Control Numbers) by Year.

- **Observation:** The trend line exhibits noticeable fluctuations over time. Crime involvement shows periods of gradual increase followed by short-term declines, indicating variability across different years. A relative peak is observed during the middle years, followed by a decline toward the most recent period, suggesting possible improvements in enforcement measures or reporting patterns.
- **Business Insight:** Understanding long-term crime trends enables policymakers and law enforcement authorities to evaluate the effectiveness of past crime control strategies and

identify periods requiring enhanced intervention. Such insights support data-driven planning and long-term policy formulation.

Visualization: A Clustered Column Chart was used.

- **X-Axis:** Year
- **Y-Axis:** Count of Incidents

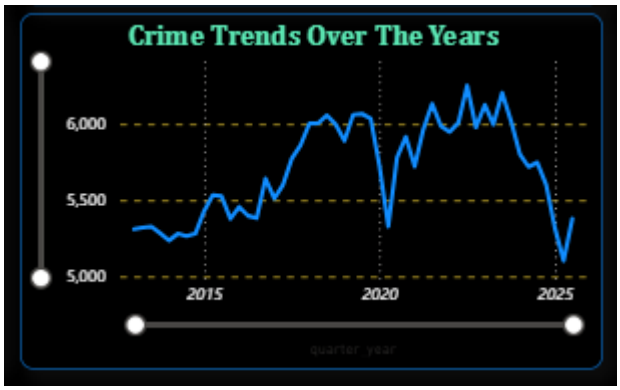

Figure 4

4.4 Objective 2: Count of Crime rate by Gender

Objective: To analyze crime involvement across different gender categories and identify gender-based patterns in reported crime data.

Analysis: The analysis was performed by aggregating the person_count measure across the person_sex_description attribute. This aggregation highlights the distribution of crime involvement among different gender groups.

Observation:

- **Males** represent the highest level of crime involvement by a significant margin.
- **Females** account for the second-largest share but remain considerably lower than males.
- **Other, Non-binary, and Unknown** gender categories contribute a minimal portion of the total crime rate.

Business Insight:

- The higher crime involvement among males may be associated with increased exposure to high-risk environments, occupational factors, and social behavior patterns.

- Gender-specific crime prevention strategies and awareness programs can help address these disparities.
- Inclusive data collection remains important to better understand emerging trends within underrepresented gender categories.

Visualization: A **Horizontal Bar Chart** was used to clearly compare crime involvement across gender categories.

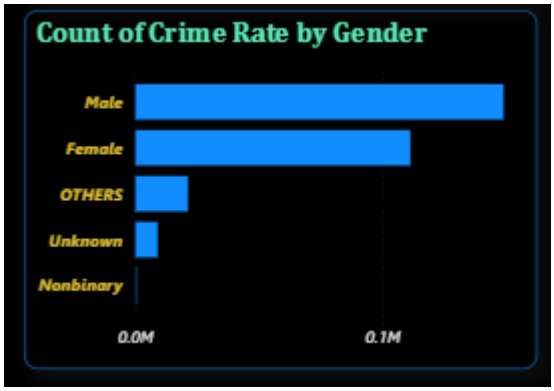

Figure 5

4.5 Objective 3: Count of Person Type by Age Range

Objective: To analyze the distribution of crime involvement across different age groups in order to identify the most affected and most involved age ranges.

Analysis: The analysis was conducted by aggregating the **person_count** field across the **age_range** attribute. This aggregation includes both victims and suspects and provides a demographic overview of crime involvement by age group.

- **Observation:**

Observation:

- The **18–29 age group** constitutes the largest share of individuals involved in reported crime incidents.
- The **30–39** and **40–49** age groups also show high levels of involvement, indicating a strong concentration within the working-age population.
- When combined, individuals between **18 and 49 years** account for the majority of total crime involvement.
- In comparison, **minors (0–17 years)** and **senior citizens (60+)** contribute a relatively smaller proportion to overall crime statistics.

- **Impact:** Higher crime involvement among young and middle-aged adults may be influenced by **socio-economic pressures**, including employment challenges, financial responsibilities, and increased urban exposure.

Visualization: A Donut Chart was used to show the percentage distribution.

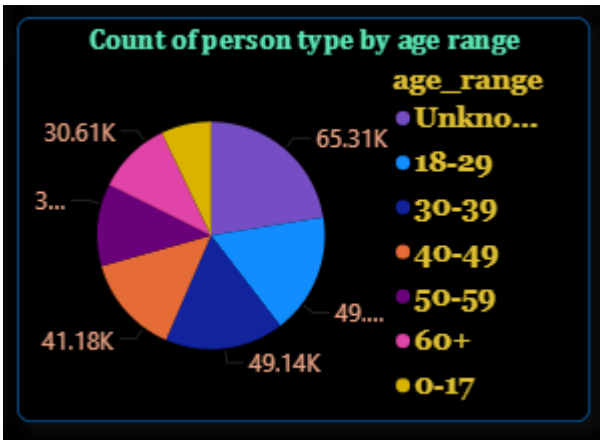

Figure 6

4.6 Objective 4: Person race_description

Objective: To analyze the distribution of crime involvement across different racial groups and identify demographic patterns within reported crime data.

Analysis: The analysis was conducted by aggregating the person count measure across the person race description attribute. This approach highlights the contribution of each racial group to overall crime involvement.

Observation:

- The **White** racial group records the highest crime involvement count.
- The **Black** and **Hispanic** groups follow closely, showing significant representation in the dataset.

.

Visualization: A **Column Chart** was used to compare crime involvement across racial categories.

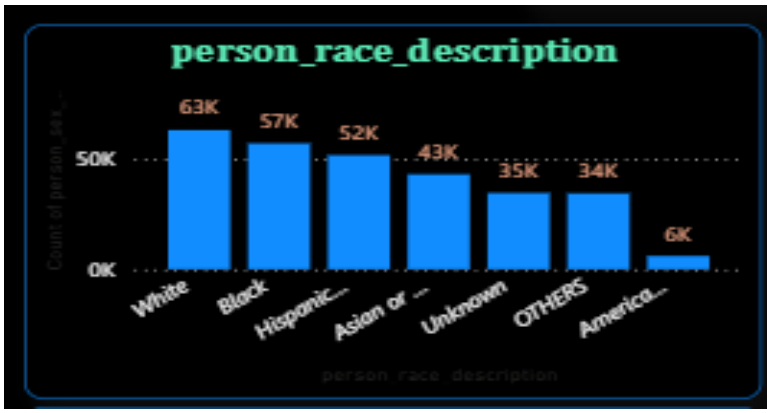

Figure 7

4.7 Objective 5: Crime category By District Description

Objective: To analyze the geographical distribution of crime involvement across different districts and understand spatial variations in crime categories.

Analysis:

The analysis was performed by plotting aggregated person count values on a geographic map using the district description field. Each bubble represents a district, with its size indicating the volume of crime involvement for a specific crime category.

Observation:

- Crime involvement is unevenly distributed across districts, with certain districts showing significantly higher concentrations.
- Urban and high-activity districts exhibit larger clusters, indicating higher crime involvement.
- Some districts display relatively low crime intensity, suggesting localized patterns of crime activity**.**

Visualization: A **Geospatial Map** visualization was used to represent crime involvement across districts. **Bubble size** corresponds to total person count, allowing quick identification of high-density crime areas.

This visual type is being retired soon. Contact your admin to...
Person crime category by district description
NORTH AMERICA
EUROPE
ASIA
Atlantic Ocean
AFRICA
SOUTH AMERICA
Indian Ocean
AUST

Crime Analytics Dashboard
District
BAYVIEW
CENTRAL
2M
Sum of...
person_type
SUSPECT
VICTIM
person_crime_category
AGGRAVATED ASSAULT
BATTERY/OTHER ASSAULT
This visual type is being retired soon. Contact your admin to...
Person crime category by district description
person_race_description
63K
57K
52K
43K
35K
34K
6K
50K
0K
White
Black
Hispanic...
Asian or ...
Unknown
OTHERS
America...
Count of person_type by person_sex_description
21.24K (7.31%)
111.2K (38.25%)
148.88K (51.21%)
person_...
Male
Female
OTHERS
Unkno...
Nonbi...
Count of Crime Rate by Gender
Male
Female
OTHERS
Unknown
Nonbinary
0.0M
0.1M
Crime Trends Over The Years
6,000
5,500
5,000
2015
2020
2025
Count of person type by age range
30.61K
65.31K
3...
49....
41.18K
49.14K
age_range
Unkno...
18-29
30-39
40-49
50-59
60+
0-17

Figure 8

5. Dashboard Design Architecture

This section explains the structural design and layout strategy followed while building the Power BI dashboard. A well-defined dashboard architecture ensures clarity, usability, and consistency across all report pages.

The dashboard follows a modular design approach where each page focuses on a specific analytical objective. Common slicers such as year, district, person type, and crime category were placed consistently across pages to maintain uniform user experience. Visual hierarchy was carefully maintained by placing key performance indicators (KPIs) at the top, followed by detailed charts and supporting visuals.

Special attention was given to color selection and contrast to ensure readability and accessibility. A dark-themed layout was chosen to reduce visual fatigue and highlight key data points effectively. Font styles and sizes were kept consistent throughout the report to maintain professional presentation standards.

This architectural approach ensures that users can intuitively navigate through the dashboard, understand insights quickly, and perform exploratory analysis without technical complexity.

Role of Slicers and Filters in Analysis

Slicers and filters play a critical role in transforming static data into interactive insights. In this dashboard, slicers were implemented to allow users to dynamically explore crime data across multiple dimensions.

Key slicers include:

- Quarter and Year
- District Description
- Person Type (Victim or Suspect)
- Age Range
- Gender
- Crime Category

These slicers enable users to isolate specific segments of data and observe how crime patterns vary across demographics and time periods. For example, selecting a particular district immediately updates all visualizations to reflect localized crime trends.

The use of cross-filtering ensures that interactions with one visual automatically update related visuals. This enhances analytical depth and supports comparative analysis across multiple variables simultaneously.

Data Modeling Strategy

A well-structured data model is essential for efficient analysis and accurate insights. In this project, a star-schema-like structure was followed to optimize performance and simplify relationships.

The central fact table contains crime incident records, while descriptive attributes such as district, gender, race, and age range function as dimension tables. Relationships were defined using appropriate cardinality to ensure correct aggregation behavior.

Proper data modeling enabled:

- Faster query performance
- Accurate filtering across visuals
- Reduced redundancy
- Scalability for future data expansion

This approach ensured that the dashboard remains responsive even when working with large volumes of data.

7.Key Performance Indicators (KPI) Analysis

Key Performance Indicators (KPIs) are measurable values used to evaluate performance and monitor outcomes against defined objectives. In the context of crime data analysis, KPIs help summarize large volumes of complex data into meaningful metrics that provide quick and actionable insights. Instead of analyzing raw crime records individually, KPIs offer a consolidated view of crime involvement across different dimensions.

In this project, KPIs were designed using aggregated values derived directly from the dataset. These indicators provide a high-level overview of crime patterns and serve as an entry point for deeper, detailed analysis through interactive visualizations.

Objective of KPI Analysis

The primary objective of incorporating KPIs in the Crime Analytics Dashboard is to:

- Provide an at-a-glance summary of overall crime involvement
- Support quick comparison across time periods and districts
- Enable decision-makers to identify high-risk areas and demographics efficiently
- Enhance dashboard readability by highlighting key metrics

KPIs act as a foundation for analytical exploration and help users quickly understand the current state of crime data before drilling down into detailed views.

KPIs Used in the Dashboard

The following KPIs were identified as relevant and meaningful based on the available dataset:

1. **Total Crime Count**

   This KPI represents the total number of individuals involved in reported crime incidents. It is calculated by aggregating the PERSON_COUNT field across all records. This metric provides a general measure of crime involvement within the selected filters.

2. **Total Victim Count**

   This KPI measures the total number of individuals classified as victims. It is derived by filtering the dataset where PERSON_TYPE = Victim and summing the corresponding PERSON_COUNT. This indicator helps identify the scale of victimization across demographics and locations.

3. **Total Suspect Count**

   This KPI represents the total number of individuals identified as suspects in reported crime incidents. It is calculated by filtering records where PERSON_TYPE = Suspect. Comparing this KPI with victim count supports comparative demographic analysis.

4. **Crime Count by Selected Time Period**

   This KPI dynamically reflects crime involvement for a selected year or quarter. It enables temporal comparison and helps identify periods with higher or lower crime activity.

5. **District-wise Crime Count**

   This KPI highlights total crime involvement within a selected police district. It supports geographical analysis and helps identify districts with high crime concentration.

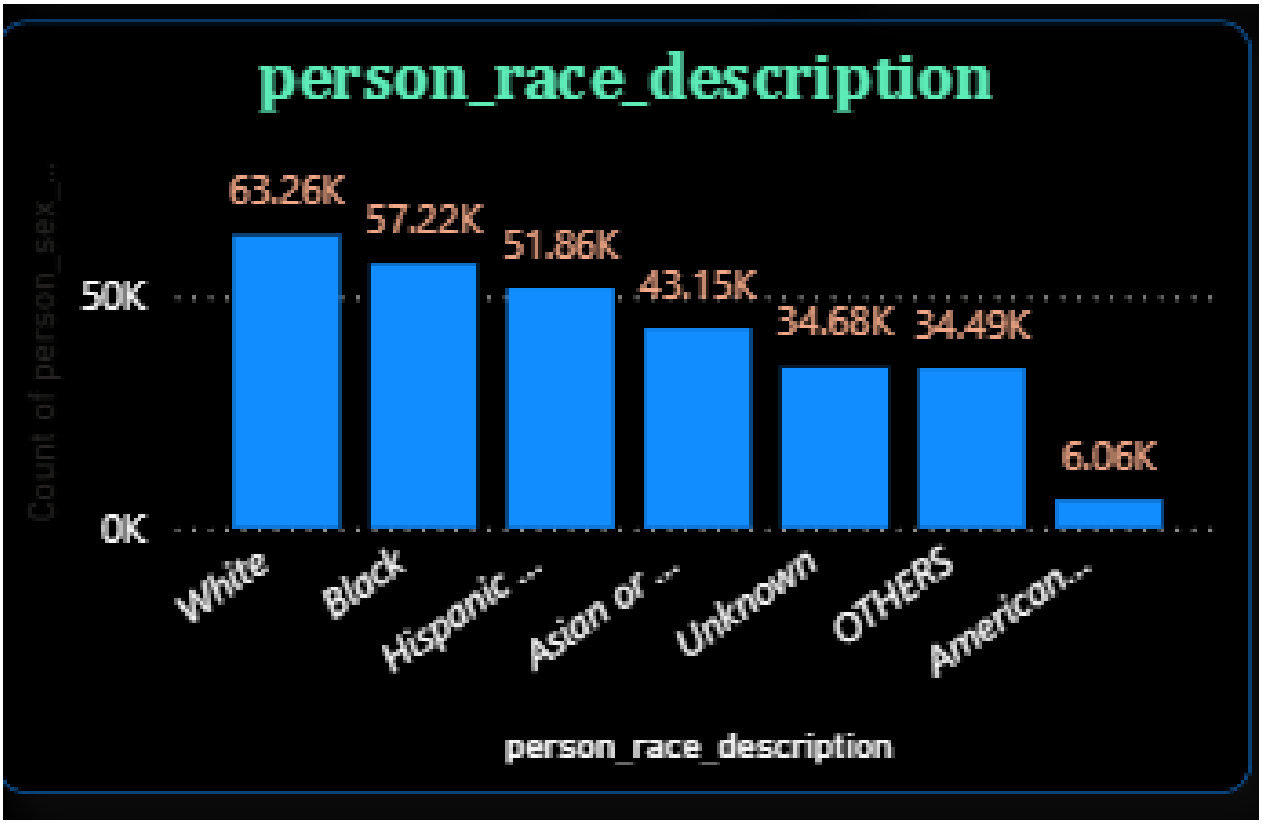

Figure 10: Crime involvement distribution across racial groups

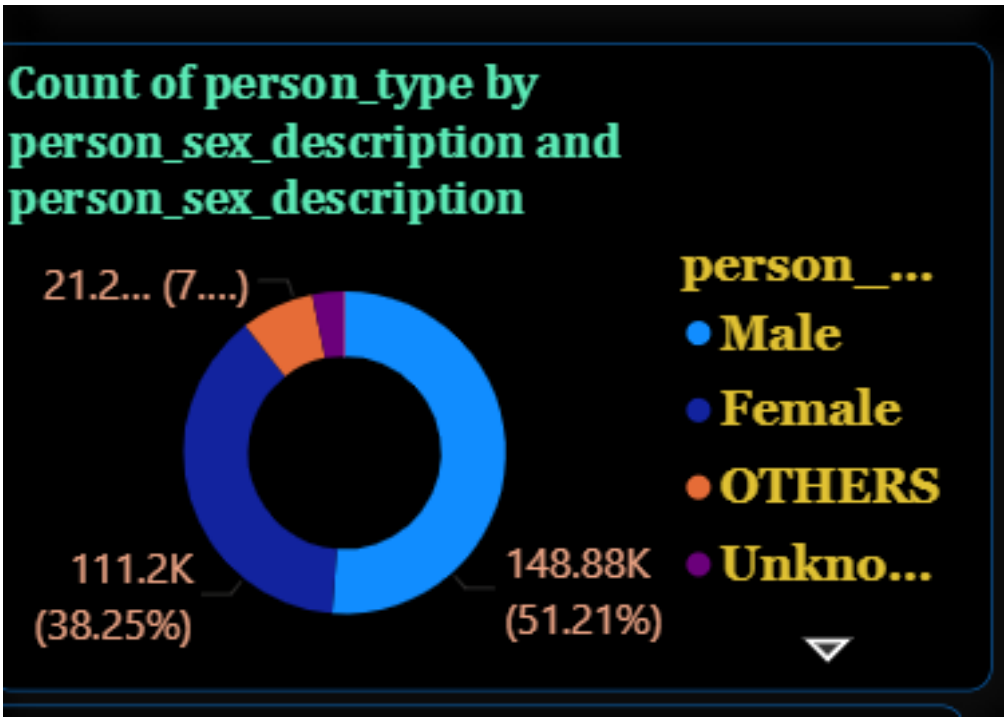

Figure 11: Gender-wise distribution of crime involvement

KPI Calculation Approach

All KPIs were created using DAX (Data Analysis Expressions) measures in Power BI. Aggregation functions such as SUM() were applied on the PERSON_COUNT field, while conditional filters were used for person type, district, and time period.

These KPIs are dynamic in nature and automatically update based on user-selected slicers and filters. This ensures consistency between KPI values and detailed visualizations across the dashboard.

Visualization of KPIs

KPIs are displayed using **card visuals** placed at the top section of the dashboard. Card visuals were selected because they present numerical values clearly and concisely, allowing users to quickly interpret key metrics without distraction.

Each KPI card is labeled appropriately to avoid ambiguity. The placement of KPI cards follows a logical flow, ensuring that users can understand overall crime involvement before exploring specific demographic or district-level analyses.

Figure 9: **KPI card representing total crime involvement**

Insights Derived from KPI Analysis

KPI analysis provides immediate insights into:

- Overall scale of crime involvement
- Comparison between victim and suspect counts
- Identification of high-activity time periods
- Recognition of districts with higher crime concentration

These insights help guide users toward deeper investigation and support data-driven decision-making.

Role of KPIs in Decision Making

By summarizing complex crime data into simple, interpretable metrics, KPIs enhance the effectiveness of the dashboard as a decision-support tool. Law enforcement agencies,

policymakers, and analysts can use KPI insights to prioritize areas requiring attention, allocate resources effectively, and monitor changes over time.

The inclusion of KPIs improves analytical efficiency and strengthens the overall usability of the Crime Analytics Dashboard.

Use of DAX Measures and Calculations

DAX (Data Analysis Expressions) played a crucial role in enhancing analytical capabilities within the dashboard. Custom measures were created to calculate aggregated values, percentages, and comparative metrics.

Examples of DAX usage include:

- Total Crime Count
- Crime Percentage by Demographic Group
- Year-over-Year Comparison Metrics
- Conditional calculations for filtered contexts

By using DAX measures instead of calculated columns wherever possible, the report maintains better performance and flexibility. These measures dynamically adjust based on slicer selections, enabling real-time insight generation.

Performance Optimization Techniques

To ensure smooth dashboard performance, several optimization techniques were implemented during development. Large datasets can significantly impact report responsiveness if not handled efficiently.

Optimization strategies included:

- Removing unnecessary columns during data loading
- Limiting visual clutter by avoiding excessive visuals on a single page
- Using measures instead of calculated columns
- Applying appropriate data granularity

These steps resulted in faster load times, reduced memory usage, and improved overall user experience**.**

Data Security and Privacy Considerations

Crime data often contains sensitive demographic information, making data security and privacy a critical concern. Although this dataset is publicly available, ethical considerations were maintained throughout the analysis.

No personally identifiable information (PII) was used in the dashboard. Data was analyzed only at an aggregated level to ensure anonymity. Visualizations were designed to present trends rather than expose individual-level details.

This responsible approach ensures compliance with ethical data usage practices and promotes trust in analytical outcomes.

Limitations of the Study

Despite providing meaningful insights, the project has certain limitations that must be acknowledged.

- The dataset is limited to reported crimes, which may not represent actual crime occurrences.
- Some demographic fields contain “Unknown” values, which can affect accuracy.
- The analysis is descriptive rather than predictive in nature.
- External socio-economic factors are not included in the dataset.

Recognizing these limitations helps in correctly interpreting the results and identifying areas for improvement.

Practical Applications of the Dashboard

The Crime Analytics Dashboard can be effectively used by multiple stakeholders:

- Law Enforcement Agencies: To identify high-risk demographics and districts.
- Policy Makers: To design targeted crime prevention programs.

- Researchers: To study demographic crime trends.
- Urban Planners: To assess safety concerns in different districts.

These applications demonstrate the real-world relevance of data analytics in public safety and governance.

Learning Outcomes of the Project

This project contributed significantly to skill development in data analytics and business intelligence.

Key learning outcomes include:

- Hands-on experience with Power BI tools
- Practical understanding of ETL processes
- Data modeling and relationship management
- Visualization best practices
- Analytical thinking and interpretation of insights

The project strengthened both technical and analytical competencies essential for a data-driven career.

Alignment with Course Objectives

The project aligns closely with the objectives of the INT374 course by applying theoretical concepts to a real-world dataset. It demonstrates the use of data visualization techniques, analytical reasoning, and dashboard storytelling.

By integrating classroom knowledge with practical implementation, the project reinforces experiential learning and industry relevance.

Scope for Academic and Industry Extension

The dashboard framework developed in this project can be extended for academic research as well as industry applications. Researchers can use the model to perform comparative studies, while organizations can adapt it for internal crime monitoring systems.

Such extensibility highlights the scalability and adaptability of Power BI-based analytics solutions.

Visualization Selection Rationale

Selecting the right visualization is crucial for conveying insights effectively. Each chart type used in this Power BI dashboard was carefully chosen based on the nature of the data and the analytical objective.

Bar and column charts were used where direct comparison between categories was required, such as gender-wise and race-wise crime involvement. These charts allow easy comparison and quick identification of dominant categories. Donut charts were selected to represent proportional distributions, especially for age group analysis, as they clearly show percentage contribution to the total.

Geospatial map visuals were chosen for district-level analysis to provide spatial context and enable intuitive understanding of geographical crime concentration. Line and trend-based visuals were used for time-series analysis to highlight changes over years and quarters.

This thoughtful selection of visuals ensures clarity, reduces misinterpretation, and enhances storytelling through data.

Dashboard Usability and User Experience (UX)

User experience plays a vital role in the effectiveness of any analytical dashboard. The design of this Power BI report prioritizes ease of use, clarity, and logical flow of information.

Navigation between report pages is straightforward, with each page dedicated to a specific objective. Consistent layout patterns help users quickly understand where to look for filters, KPIs, and insights. Interactive elements such as slicers, tooltips, and cross-highlighting improve engagement and exploration.

The use of minimal text on visuals ensures that users focus on insights rather than clutter. Tooltips provide additional context without overwhelming the dashboard interface. Overall, the

user-centric design ensures that both technical and non-technical users can interpret insights with ease.

Impact of Interactive Analytics on Decision Making

Interactive dashboards significantly enhance decision-making capabilities compared to traditional static reports. This project demonstrates how interactive analytics empowers stakeholders to explore data dynamically and uncover insights relevant to their specific needs. Users can instantly drill down into specific districts, demographic groups, or time periods without requiring technical expertise. This flexibility allows law enforcement officials and policymakers to test assumptions, compare scenarios, and validate hypotheses in real time.
By enabling rapid insight generation, interactive analytics reduces decision latency and improves responsiveness in crime prevention and strategic planning efforts.

Comparative Analysis Capabilities

One of the strengths of the Power BI dashboard lies in its ability to support comparative analysis across multiple dimensions. Users can compare crime involvement between victims and suspects, across age groups, genders, and districts.
Such comparisons help identify disproportionate patterns, emerging trends, and demographic disparities. For example, comparing age group involvement across different districts can reveal localized demographic crime trends.
This comparative capability strengthens analytical depth and supports evidence-based conclusions.

Reporting and Documentation Standards

Proper documentation is essential for academic and professional projects. This report follows a structured documentation approach, including clear objectives, methodology, analysis, results, and conclusions.

Each visualization is supported by descriptive analysis and business insights to ensure interpretability. Figures are sequentially numbered, and sections are logically ordered to maintain academic clarity.
Adhering to documentation standards enhances the credibility of the project and ensures it meets university evaluation criteria.

Ethical Use of Open Government Data

The dataset used in this project originates from an open government source, emphasizing transparency and public accessibility. Ethical responsibility was maintained throughout the project by ensuring respectful and appropriate use of data.
The analysis focuses on patterns and trends rather than individual-level interpretation. No conclusions were drawn that could lead to bias, discrimination, or stigmatization of specific demographic groups.
This ethical approach ensures that data analytics contributes positively to public understanding and policy formulation.

8. Conclusion

The **Victim and Suspect Demographic Crime Analysis Dashboard** project has successfully demonstrated the practical application of **Business Intelligence techniques** in understanding crime patterns from a demographic perspective. By analyzing structured crime data segmented by **time period, district, age group, gender, race, and person type**, the project transformed raw records into an interactive, insight-driven decision support system.
The Power BI dashboard enabled comprehensive analysis of demographic involvement in crime and provided meaningful visual interpretations to support data-driven policy formulation and law enforcement planning.
Key Conclusions:

1. **Data Integration:** The demographic crime dataset was successfully cleaned, standardized, and transformed into a unified analytical model, ensuring reliable and consistent analysis across multiple dimensions such as time, location, and demographics.
2. **Strategic Insight:** Analysis revealed that **young and middle-aged adults (18–49 years)** account for the majority of crime involvement, highlighting the importance of targeted interventions for this population segment.
3. **Gender Based Pattern :** The findings indicate that **males show significantly higher crime involvement** compared to other gender categories, while females represent the second-largest group. Other and unknown gender categories contribute a comparatively smaller share..
4. **Spatial Awareness through Visualization::** District-wise and map-based visualizations proved highly effective in identifying **geographical concentrations of crime involvement**, allowing stakeholders to quickly recognize high-activity areas.

This project serves as a prototype for how modern data analytics can enhance the efficiency and responsiveness of law enforcement agencies.

9. Future Scope

While the current dashboard is comprehensive, there are several avenues for future enhancement:

1. **Real-Time API Integration:** Future versions of the dashboard can integrate live data sources or APIs to enable real-time or near-real-time crime monitoring, reducing reliance on static datasets.
2. **Predictive Analytics:** Machine Learning models implemented using **Python or R** can be incorporated to forecast future crime trends and identify high-risk demographic segments.
3. **Social Time data Integration :** Integrating census and socio-economic indicators such as income levels, employment rates, and population density can help uncover underlying drivers of demographic crime patterns.

10. References

Dataset Source: https://catalog.data.gov/dataset/police-department-reported-victim-and-suspect-demographic-data

1. **Microsoft Power BI Documentation:** Official guides on DAX and Power Query.

- URL: https://learn.microsoft.com/en-us/power-bi/

2. **Lecture Notes:** INT374 Data Visualization and Analytics, Lovely Professional University.
3. **Data Visualization Best Practices:** "Storytelling with Data" by Cole Nussbaumer Knaflic (General Reference).
4. Google Data Analytics Professional Certificate – Course Materials on Data Cleaning and Visualization.

11. Social Media Engagement

Link:

https://www.linkedin.com/posts/jyotikushwaha439_powerbi-crimeanalytics-dataanalytics-activity-7408925545096392704-ci3X?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEf3rNcBo0sHyfuPzhX7dU5XLHesQ9KdoA4

Figure 9